MILITARY SPECIFICATION

GATE TRANSMIT, BOARD B: 10549207

1, SCOPE

1.1 This specification covers the transmit gate, board B, used in assembly 2A5 of Receiver- Transmitter, Radar RT-860/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-G-50672	Gate, Transmit Board A: 10549205
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

Fsc 1285

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
M1L-STD-81O	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F105492O7	Transmit Gate, Board B
FI0549203	Transmit Gate Schematic
c10549172	Delay Line
D10548227	Transformer, Pulse
D10548199	Transformer, Pulse
F105492O5	Transmit Gate, Board A

PACKAGING DATA SHEET

10549207	Packaging of Transmit Gate, Board B

(Copies of specifications, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. - The transmit gate, board B, shall be manufactured in accordance with Drawing F1054 9207 and drawings pertaining thereto.

3.2 General specifications. - Where applicable, the transmit gate, board B, shall meet the requirements of MIL-F-13926 and MIL-C46357 as follows :

3.2.1 MIL-F-13926.

a. Order of Precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2. MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. - The transmit gate, board B, shall be mounted on the vibration table in normal service position. The transmit gate, board B, shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 55 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed l-Hz per second. Vibration test shall be conducted under ambient room conditions.

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3.3.2 Shock. - The transmit gate, board B, shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ± 1 milliseconds (ins). The G value of each impulse shall be within ± 10 percent when measured with a 0.2 and 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3. 3.1 Operating. - The transmit gate, board B, shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg.)

3. 3.3.2 Nonoperating - The transmit gate, board B, shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3. 3.4 Temperature.

3. 3.4.1 Operating. - The transmit gate, board B, shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40” F and -155° F.

3.3.4.2 Nonoperating (storage). - The transmit gate, board B, shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65° F and +160° F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3. 3.5 Humidity (nonoperating). - The transmit gate, board B, shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ± 5 percent for a period of not less than 10 days.

3.4 Electrical. - Each conductor on the transmit gate, board B, shall be electrically continuous as specified on the applicable portions of schematic diagram F105492O3.

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3.5 Performance. - The transmit gate, board B, shall yield the output voltages and signals as specified in Table I with the following conditions.

3. The input voltages and signals as specified in Table I.
b. The transmit gate, board B, shall be connected to its associated circuitry as specified on schematic diagram F105492O3.

TABLE I INPUT AND OUTPUT CHARACTERISTICS

REQUIREMENT	CHARACTERISTICS	TERMINALS
Input voltages	+15. O ± 0. 1 Vdc; 1.0 mV maximum ripple	E21, E2S(+) E20 (return)
	-15.0 ± O. 1 Vdc; 1.0 mV maximum ripple	E19(-) E20 (return)
Input. signals	Sine wave; gated on for 1.8 ± 0 .2 µs. A half-cycle width of 280 ± 10 ns with an amplitude of 5 ± 1 V peak-to-peak.	E17
	Same as signal at El 7 except 180 degrees out of phase.	**E18**
	Pulse: 0.5 ± .05 µs wide, +8 ± 2 V pcak-to-peak amplitude at 20.0 ± O. 1 kHz.	**E23**
	Same as signal at E23 except of opposite polarity.	E24
	Sinewave: 60 mHz; amplitude, O dB into a 50-ohm load.	E27
Output voltages	5.1 Vdc, ± 5% Zener regulated zero to 5.6 ± 5% Vdc, variable	E22 E25
Output signal	Cosine squared waveform as shown in Figure 1.	E26

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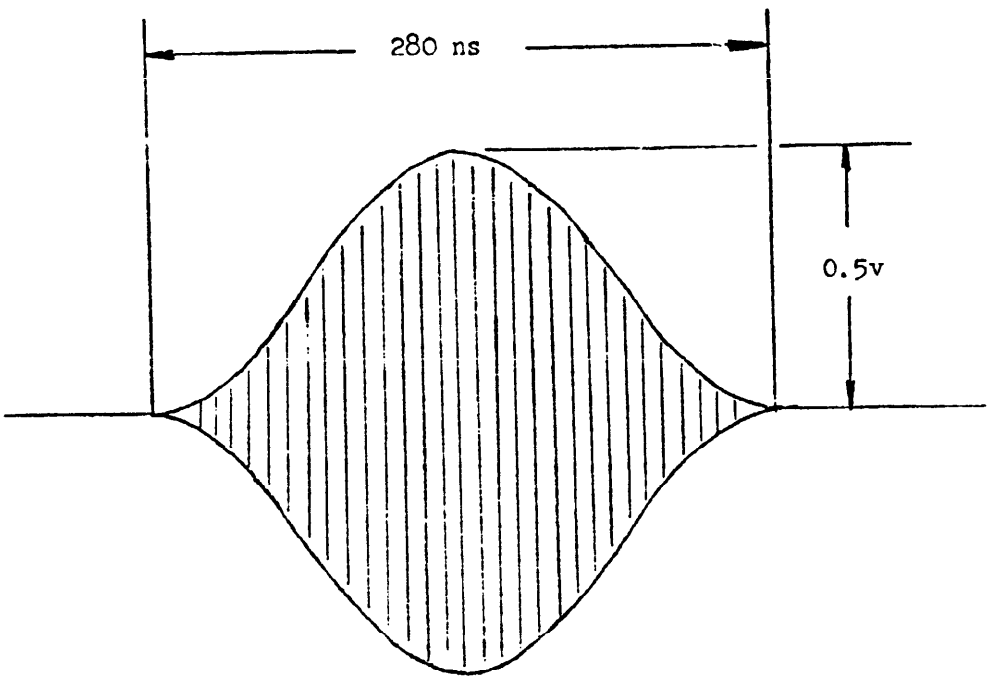


FIGURE 1. TRANSMIT GATE, BOARD B, OUTPUT SIGNAL

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4. QUALITY ASSURANCE PROVSIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise spccified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services con - form to prescribed requirements.

4. 1.1 General provisions. - The component and subassembly inspec- tion requirements of MIL-F-13926 form a part of the quality assurance pro- visions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) submission. - The requirement for first article approval and responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three transmit gates, board B manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it con- forms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract.. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Govcrnment reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2, 2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD- 105 and the sampling plans specified in 4. 3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P- 14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4. 3.2.3 Functional tests. -1 he requirements and tests in Table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60° F and 90° F),

TAB LE II. - CLASSIFICATION OF DEFECTS
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3*5	4.6.3
MINOR: NONE		

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4. 3.3 Acceptance and rejection. - Rejected lots shall be screened for All defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One transmit gate, board B, shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table 111. Three transmit gates, board B shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table IV.

TABLE III. - SPECIAL SAMPLING

	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - Visual
302.	General specifications	3.2	MIL-13926 & MIL-C-46357- Visual

TABLE IV. - SPECIAL SAMPLING, ENVIRONMENTAL

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude	3. 3.3	4.6.1.3
306.	Temperature	3. 3.4	4.6.1.4
307.	Humidity - nonoperating	3.3.5	4.6. 1.5

4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3. 3).

4.5 Inspection Equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-1-45607.

4. 5.1 Government-fumished inspection equipment. - Where the contract provides for Government -furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4. 5.2 Contract-furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4. 5.2.2 Contractor desire. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedure specified in "Examination" and "Test Facilities" requirements of MIL- F- 13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test Equipment. - In conjunction with 4.5.2.2, the standard test equipment listed in Table VI shall be utilized in performing the applicable test of 4.6.

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TABLE VI. - TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	Power Supply	Dual output: ± 15 Vdc; less than 1. 0 mV ripple; 300 mA capacity.
2.	Pulse generator	Pulse width: 0.5 µs. Rise time: less than 15 µs. Pulse amplitude: -10 volts across 50-ohm load. PFR: 20 kHz.
3.	Rf signal generator	Frequency: 60 MHz. Amplitude: O dB across 50-ohm load, Harmonic output: 30 dB below the carrier. Spurious AM: 70 dB below the carrier,
4.	Digital voltmeter	Range: O to 50 Vdc. Accuracy: +0. 1 percent of reading, ± 1 digit.
5.	Oscilloscope	Dual channel. Vertical bandwidth: dc to 150 MHz. Vertical rise time: 2.4 ns. Vertical sensitivity: 20 mV to 10 V/cm. Horizontal time base: 0.05 µs/cm to 5 s/cm.
6.	Termination	50 ohm ± 1 percent, 1/2 W, O to 500 MHz.
7.	Test Harness	As specified in Figure 2 See Table VII for test harness parts list.

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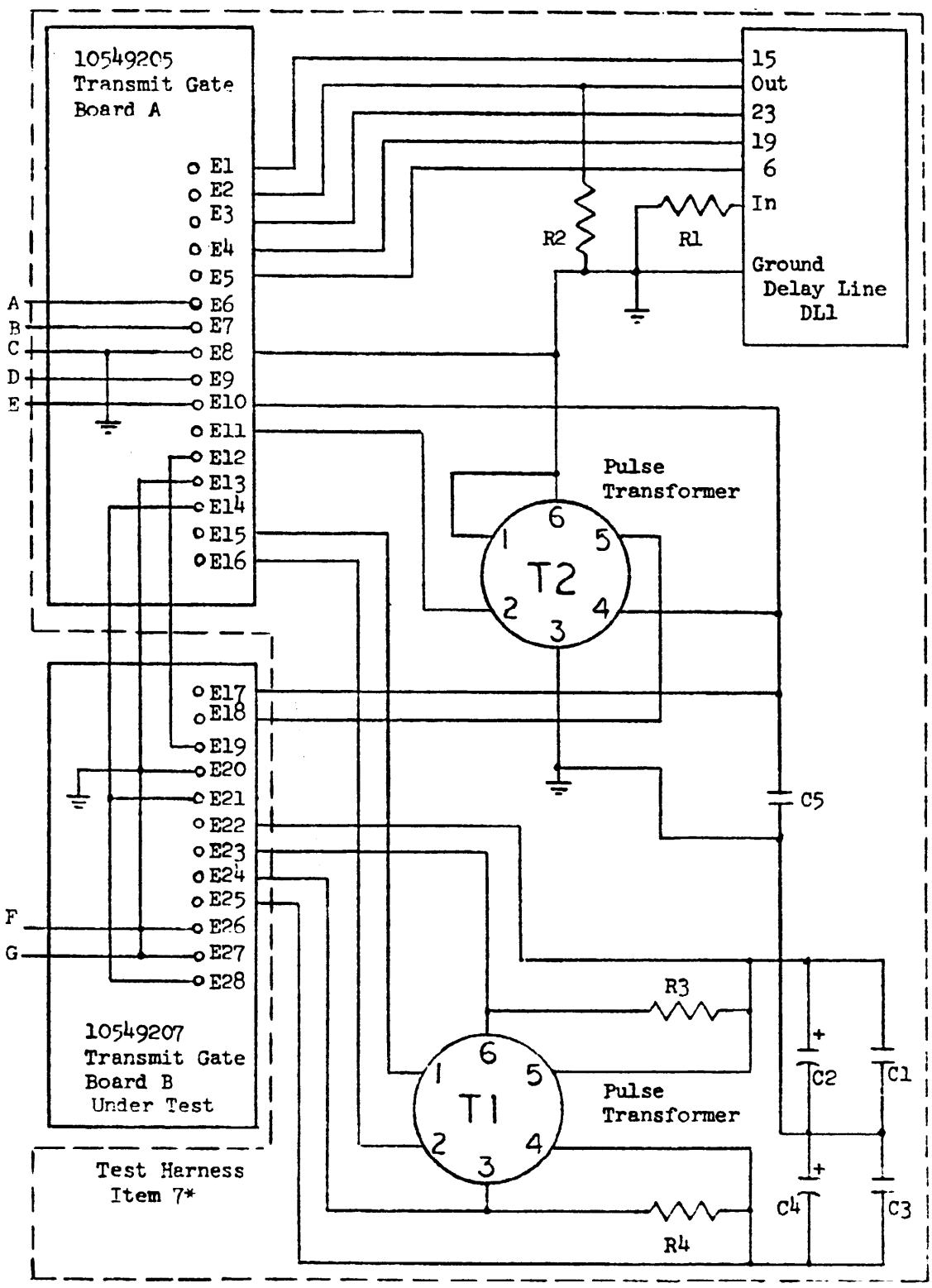
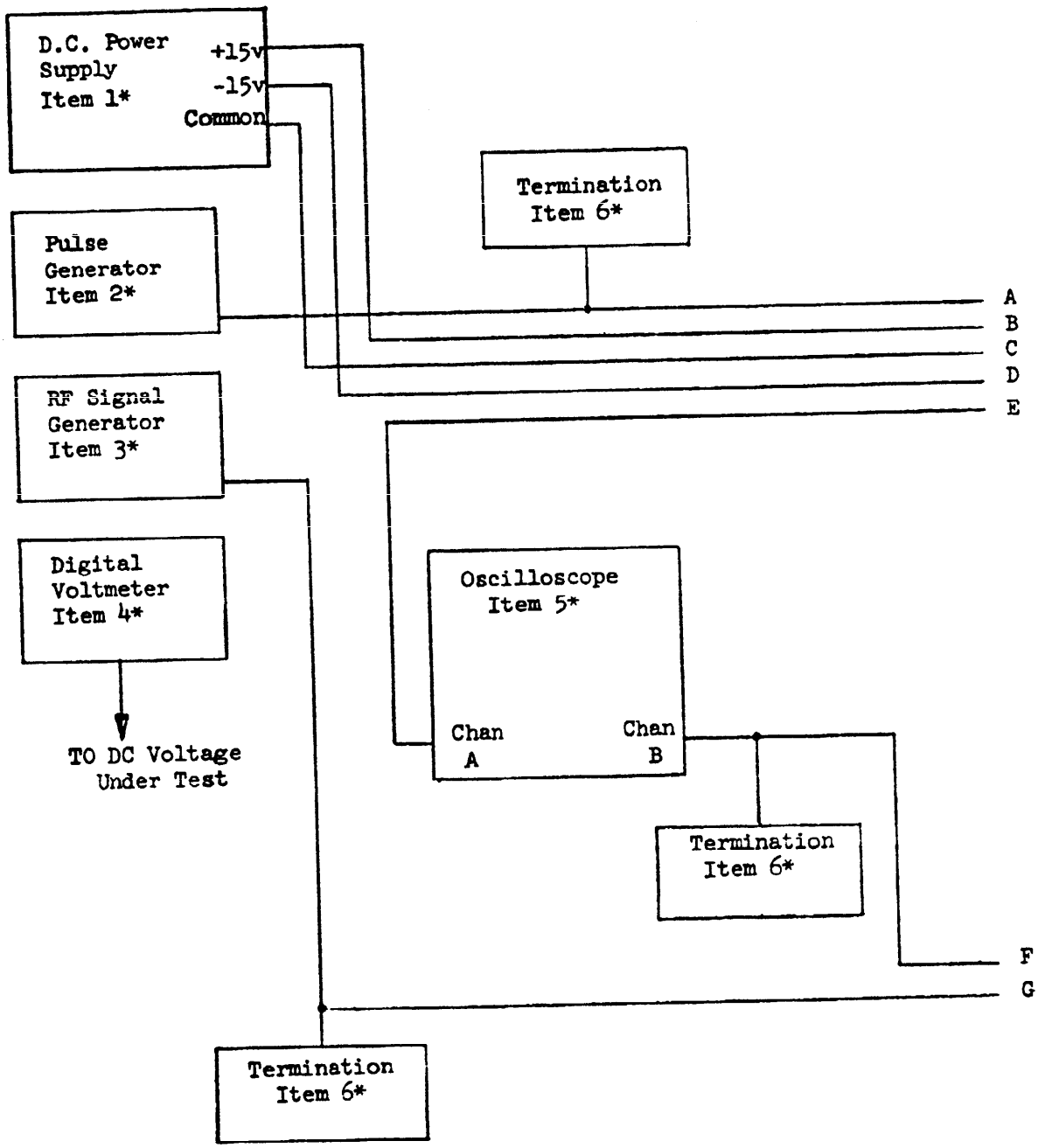


FIGURE 2. TRANSMIT GATE, BOARD B TEST SETUP (Sheet 1 of 2)

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*Refer to 4.5.2.3

FIGURE 2. TRANSMIT GATE, BOARD B TEST SETUP (Sheet 2 of 2)

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TABLE VII. - TEST HARNESS PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
DL1	Delay line 10549172.
T1	Transformer, pulse 10548227.
T2	Transformer, pulse 10548199.
"A"	Transmit gate 10549205 (board A).
RI, R2	Resistor, fixed, composition, 150 ohm ± 5%, 1/4 W.
R3, R4	Resistor, fixed, composition, 470 ohm ± 5%, 1/4 w.
c1, C3	Capacitor, fixed, mica, 560 pF ± 1O%, 100 WVdc.
C5	Capacitor, fixed, mica 68 pF ± 1O%, 200 WVdc.
C2, C4	Capacitor, fixed, electrolytic, 25 µF ± 20%, 8 WVdc.

NOTE: Unless otherwise specified in Figure 2, all interconnecting coaxial cables shall be RG-58/U; .

4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-81O except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MTL-STD-81O.

4.6. 1.1 Vibration. - The test shall be performed as specified in Method 514, procedure IX, Part T of MIL-S TD-81O except that failure criteria do not apply. After vibration, the transmit gate, board B, shall be tested as specified in 4. 6.2 and 4. 6.3 to determine compliance with 3.3.1.

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4.6. 1.2 Shock. - This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms *and* 15 G. After the shock impulses, the transmit gate, board B, shall be tested as specified in 4.6,2 and 4.6.3 to determine compliance with 3.3.2.

4. 6.1.3 Altitude (nonoperating and operating). - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-81O except that in Step 2 the altitude shall be limited to 40,000 feet (5.5 inches Hg). Operation of Step 3 shall be the performance tests specified in 4.6.3.

4.6. 1.4 Temperature.

4.6. 1.4.1 High (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The tem - perature of Step 2 shall be +160° F. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be -155° F. In Step 5 the temperature shall be maintained for 4 hours and the transmit gate, board B shall be tested in accordance with 4. 6.2 and 4.6.3 during this time.

4. 6.1.4.2 Low (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-81O. The temperature of Step 2 shall be -65° F maintained for not less than 4 hours. The temperature of Step 4 shall be -40° F. In Steps 5 and 7, the transmit gate, board B shall be tested in accordance with 4.6.2 and 4.6.3.

4.6. 1.5 Humidity (nonoperating. - This test shall be performed in accordance with Procedure H, Method 507 of MIL-STD-810. The transmit gate, board B, shall be tested in accordance with 4.6.3.

4.6.2 Electrical - Continuity shall be inspected by standard rneasuring equipment (SME) and visual examination for compliance with 3.4.1.

4. 6.3 Performance. - The selected transmit gate boards shall be tested as follows: Connect test equipment, items 1 through 7 to transmit gate, board B, as shown in Figure 2. Adjust power supply, item 1; pulse generator, Item 2; and fr signal generator to deliver the input voltages and signals as specified in Table I. Use oscilloscope, item 5, to observe each of the output signals as specified in Table I. Use the digital voltmeter, item 3, to observe the output voltages as specified in Table I. The outputs shall meet the requirements of Table I.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10549207. The level of protection shall be specified in the procurement document.

6. NOTES

6.1 Intended use. - The transmit gate, board B, is used in the transmit gate (2A5), a subassembly of Receiver-Transmitter, Radar, RT-860/VPS-2, part of Radar Set AN/VPS-2. This board consists of two balanced diode bridges used to generate the modulated rf burst.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5. 1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285 -A274

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SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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